

SRM Institute Of Science And Technology

SRM Campus Economy Platform

Subject Name : SEPM

Subject Code : 21CSC303J

Presented By :

CSE CORE D - Batch_2

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Software Project Identification

- **Project Title:** SRM Campus Economy Platform
- **System Type:** Multi-tenant, hybrid B2B (Business-to-Consumer) and P2P (Peer-to-Peer) digital marketplace.
- **Core Architecture:** Decoupled client-server model (Web/Mobile Frontend connected to an existing asynchronous API Gateway and real-time NoSQL database).
- **Target Environment:** SRM University ecosystem.

The Business Case: Formalizes moderated campus commerce to unlock hyper-local economic efficiency using "Asymmetric Permissions."

Targeted Value:

For Students: A secure, exclusive trading and freelance hub (strictly verified via @srmist.edu.in credentials).

For Verified Shops: A dedicated digital storefront to increase order capacity and connect directly with the student body.

PROBLEM STATEMENT

Current Challenges in Campus Commerce

The Core Issue: Campus commerce is currently highly fragmented, relying heavily on disorganized WhatsApp groups and generic off-campus apps.

Critical Pain Points:

High Noise-to-Signal Ratio: The student experience is severely degraded by constant spam from unverified local businesses.

Lack of Trust & Security: Without a centralized verification system, buyers cannot confirm if they are dealing with an actual student or an officially sanctioned campus shop.

Inefficiency: Chaotic and unstructured interactions throttle the natural economic potential of the university ecosystem.

Stakeholders & Users Title: Platform Ecosystem & Asymmetric Permissions

Verified Students (Consumers & Peers)

- Gated by @srmist.edu.in credentials.
- Can purchase from shops (B2C) and trade/freelance in exclusive student-only zones (P2P).

Verified Shops (B2C Vendors):

- Local campus businesses requiring manual admin approval.
- Manage digital storefronts and process orders, but are restricted from posting in P2P zones.

System Administrators (Overseers):

- University staff with root-level access.
- Approve shop applications, moderate content, and maintain platform security.

SRM Campus Economy Platform – SRS Summary

Objective: Develop a multi-tenant, role-based digital marketplace for the SRM University ecosystem, supporting peer-to-peer (P2P) student trading and business-to-consumer (B2C) transactions with verified campus shops.

Architecture: Decoupled client-server model utilizing a FastAPI (Python) asynchronous backend, a real-time Firebase NoSQL database, and responsive frontend clients (Web/Mobile via React/Next.js or Flutter).

Core Security: A strict "5-Door Bouncer" security architecture for role-based access control, enforced by FastAPI dependency injection validating JWT/OAuth tokens from Google Cloud Identity Platform.

Key Functional Modules: Exclusive @srmist.edu.in domain-verified authentication, CRUD product/service listings, business verification & dashboard, cart/checkout with inventory validation, real-time negotiation engine (chat/bidding with sync), and an Admin "God Mode" for role elevation and global moderation.

Non-Functional Standards: Scalable async I/O with REST responses under 200ms, secure secret management, native graceful error handling (HTTP 400), modular backend code (APIRouter), and an automatically generated OpenAPI contract.

Decoupled 3-Tier Serverless Architecture

Tier 1: Presentation Layer (Client Interfaces)

Platforms: Student App (Mobile), Shop Dashboard (Web), and Admin Portal.

Connectivity: Uses stateless RESTful APIs, with direct WebSockets exclusively for real-time features.

Tier 2: Application Layer (FastAPI Gateway)

"5-Door Bouncer" Security: Custom middleware that authenticates and strictly routes traffic based on user roles (Student, Shop, God Mode).

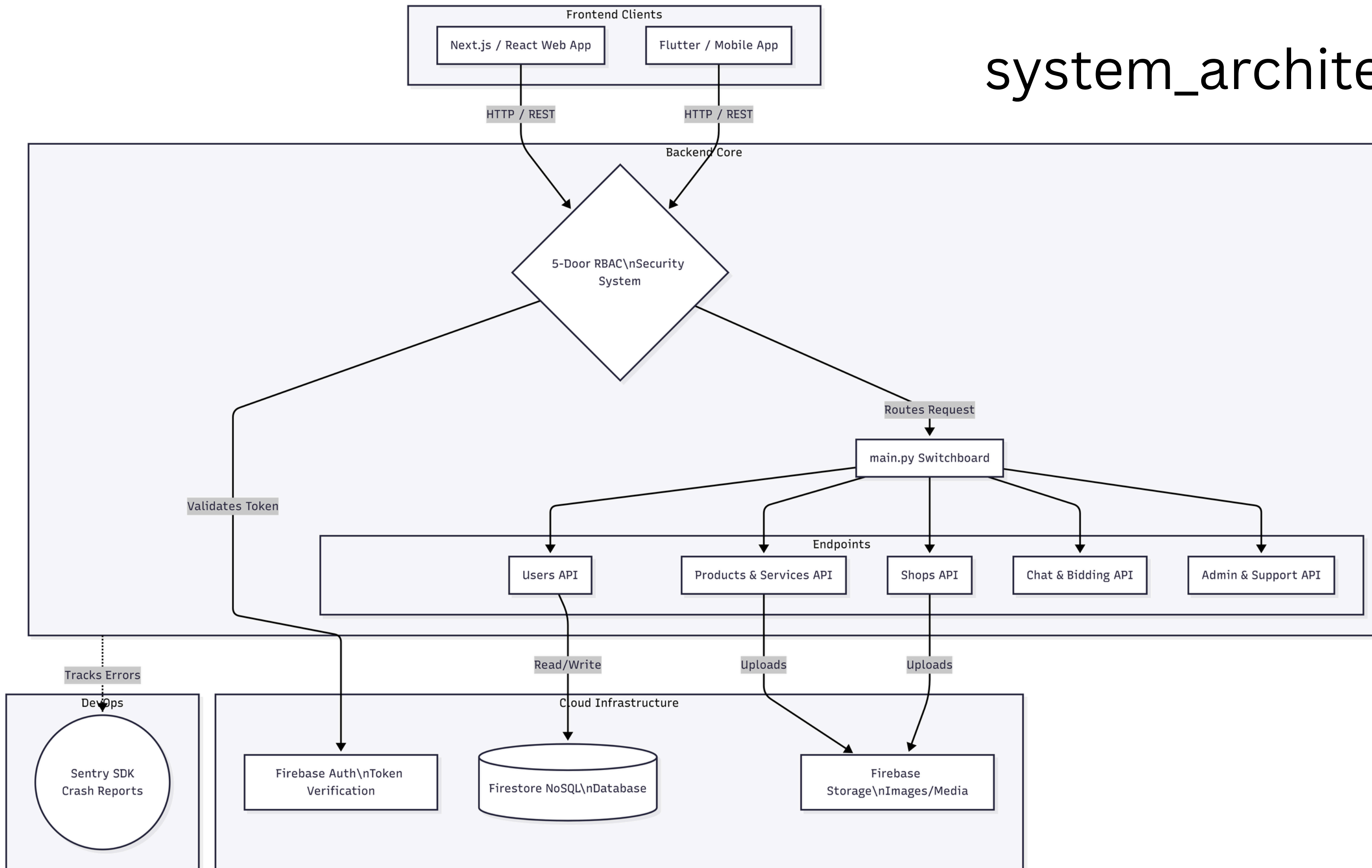
Modular Micro-services: Executes core business logic (/users, /products, /orders) and formats data payloads.

Tier 3: Data & Cloud Infrastructure (Firebase Firestore)

Primary Storage: NoSQL cloud database acting as the single source of truth.

Sub-Second Latency: Utilizes Real-time Firestore Listeners to bypass the API switchboard, syncing chat and bidding messages instantly.

system_architecture_2



Use_case_diagram

Verified Student (P2P & Consumer)

Access: Strictly gated by @srmist.edu.in Google login.

Actions: Browse global marketplace, place orders with shops, post P2P second-hand items/services, and use real-time chat/bidding.

Constraints: Completely blocked from vendor dashboards and B2C catalog zones.

Verified Shop (B2C Vendor)

Access: Requires manual role elevation by an Admin.

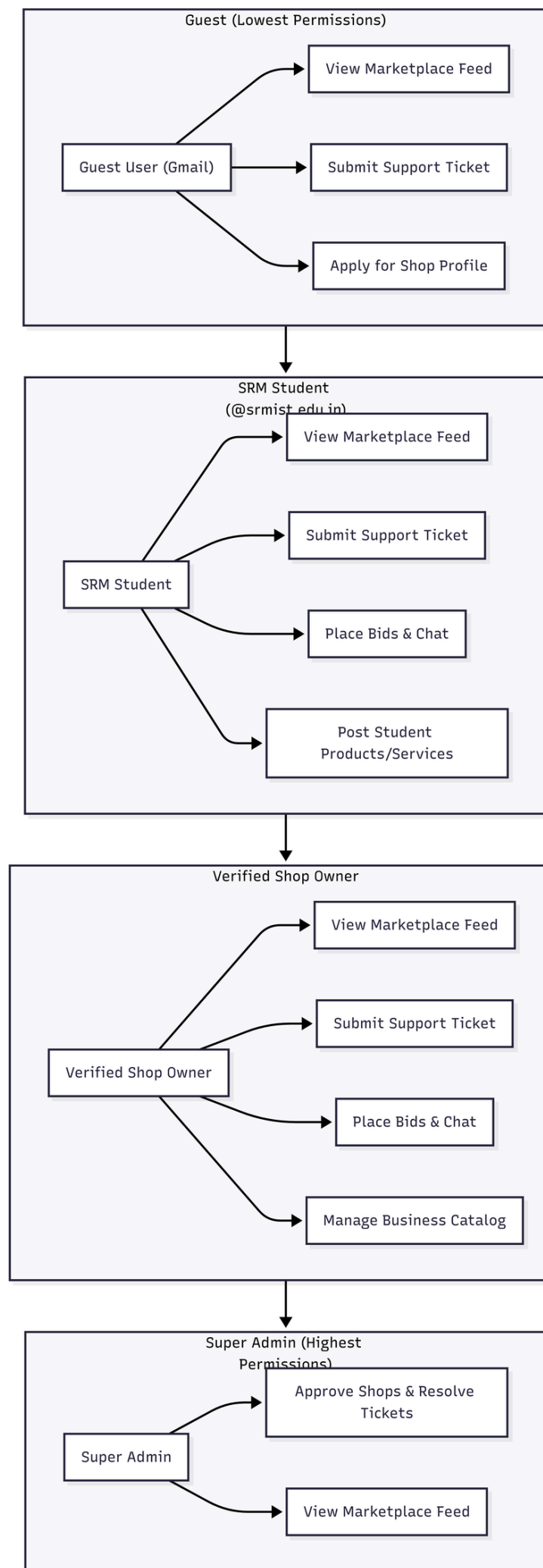
Actions: Manage digital storefronts (CRUD catalog), view live order queues, and update fulfillment statuses.

Constraints: Cannot post listings in student-exclusive P2P zones.

System Admin (God Mode)

Actions: Review and approve/reject "Shop Verification" applications.

Moderation: Suspend bad actors, instantly delete inappropriate listings globally, and view high-level system analytics.



NoSQL Database Schema (Firebase Firestore)

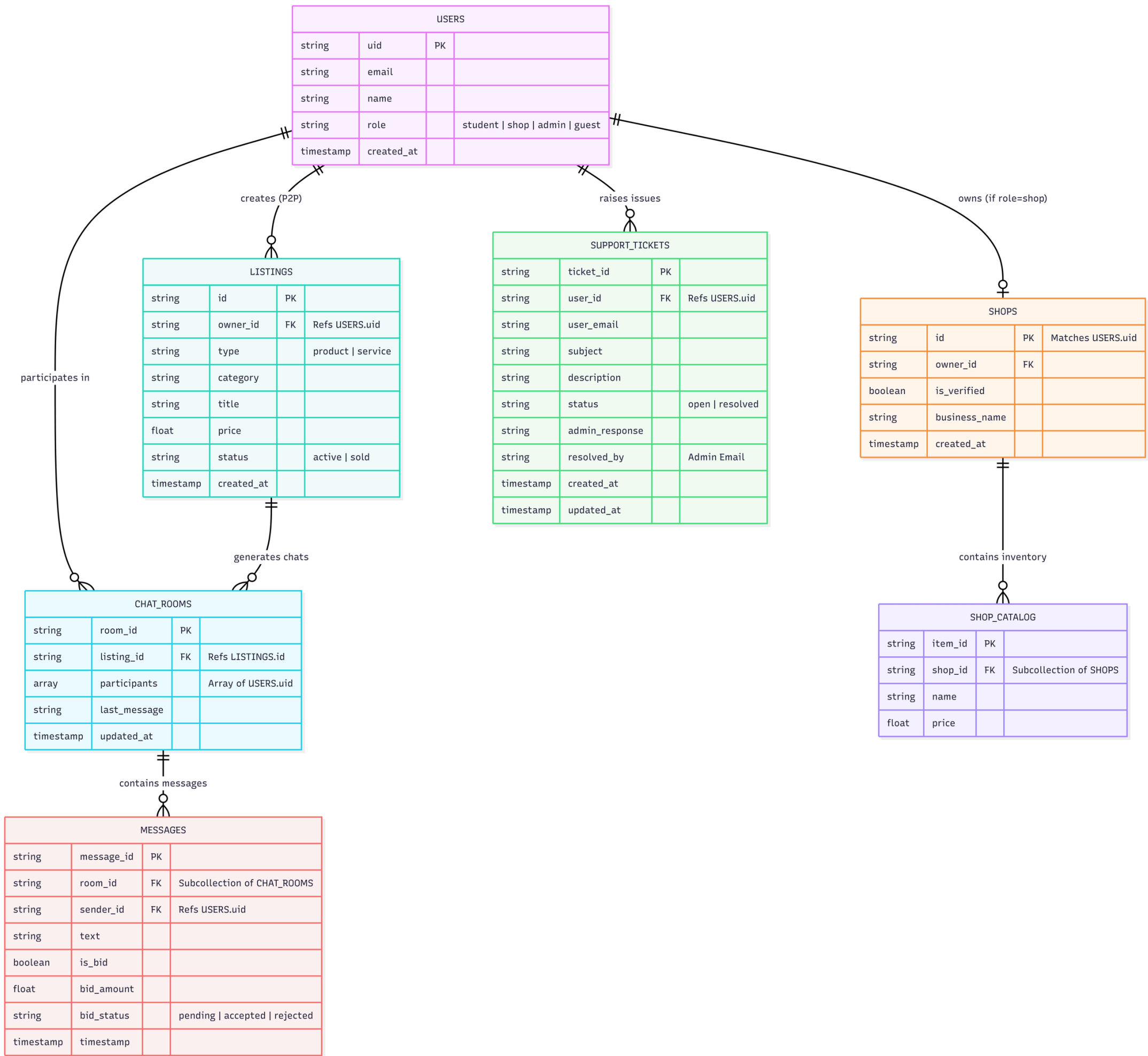
Architecture Strategy: Horizontally scalable document-collection schema, denormalized to prioritize fast read speeds over traditional SQL constraints.

users (Identity Registry): The central hub linking authentication (uid) to strict platform roles ("student", "shop", "admin").

listings (Unified Catalog): Consolidates all marketplace inventory. Uses the type field to separate B2C shop products from P2P student items and services.

orders (B2C Transactions): Manages the purchasing lifecycle connecting a student_id (buyer) to a shop_id (vendor), tracking the order_status from pending to completion.

negotiation_rooms (Real-Time Engine): Powers the P2P chat and bidding system. Links two participants to a specific listing (context_item), tracking the latest_bid and sub-collections of timestamped messages.



ER_diagram

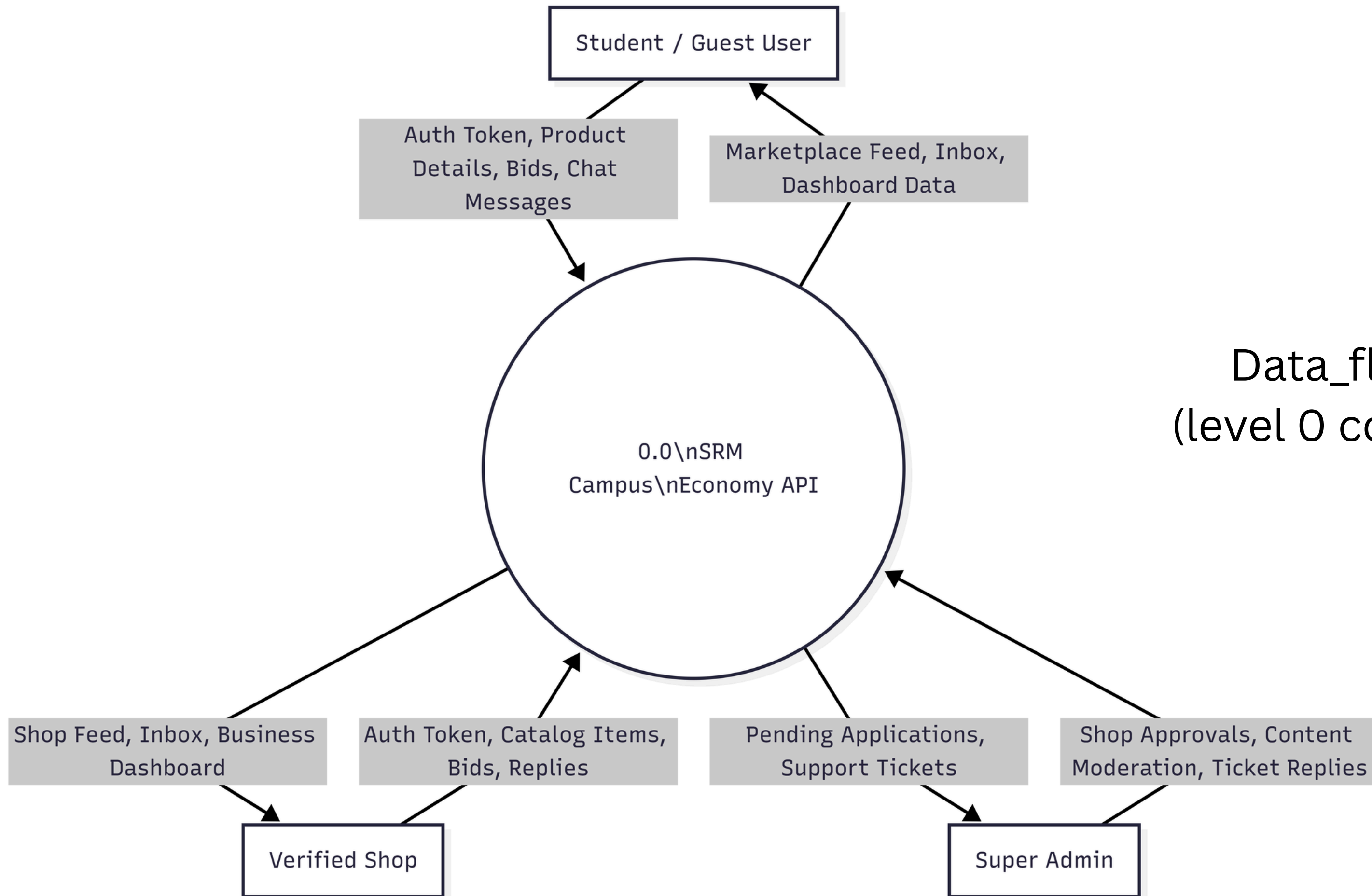
Data Flow Architecture (DFD)

Level 0: Context Diagram (External Interactions)

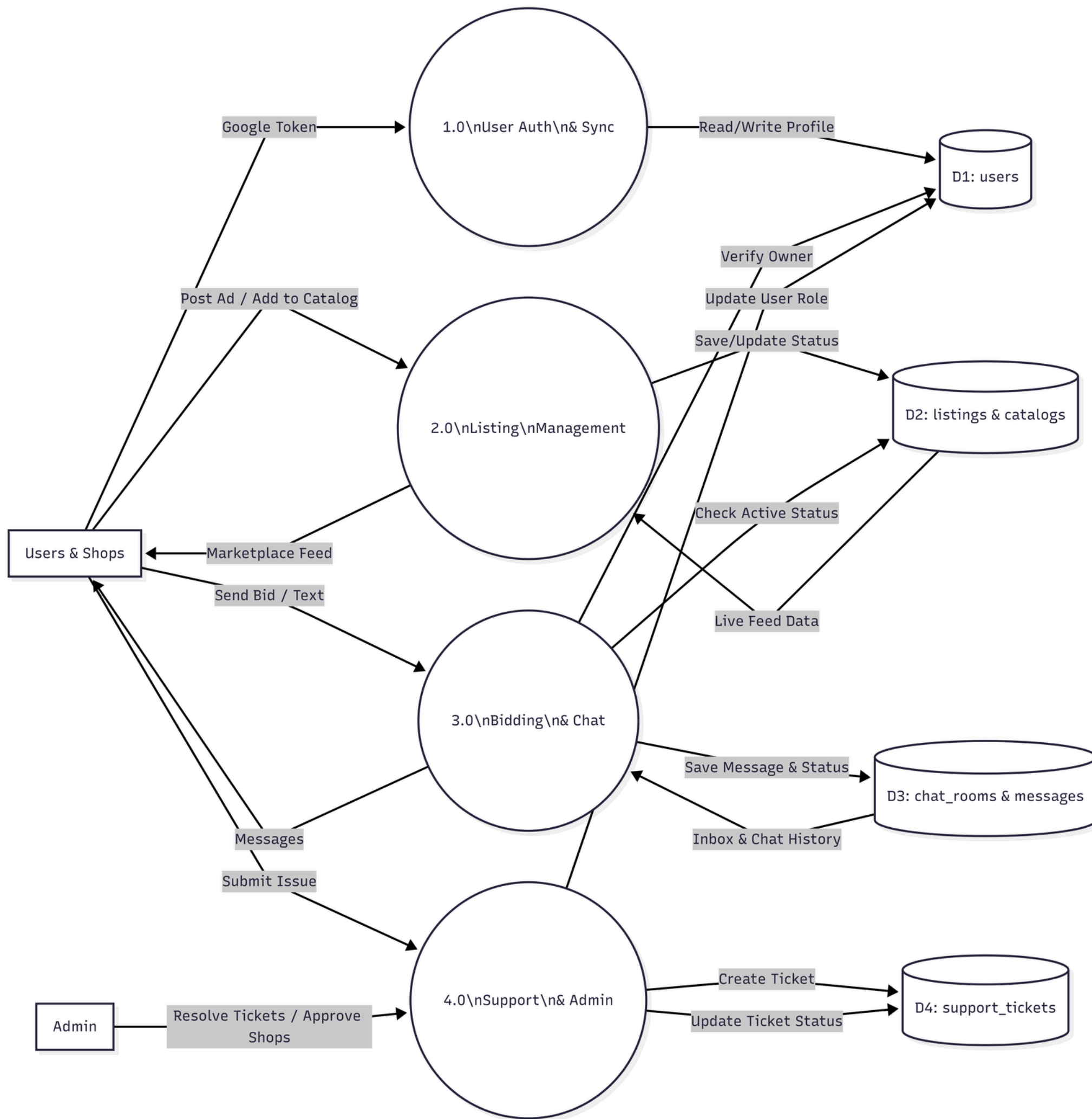
- Verified Student: Inputs P2P listings, cart payloads, and bids. Receives rendered marketplace feeds and real-time updates.
- Verified Shop: Inputs vendor applications, catalog updates, and order state changes. Receives live incoming order queues.
- System Admin: Inputs moderation commands and role approvals. Receives system logs and pending applications.

Level 1: Process Decomposition (API Micro-Services)

- 1.0 Security & Routing ("5-Door Bouncer"): Intercepts HTTP requests, decodes auth tokens, checks database roles, and routes or rejects traffic.
- 2.0 Catalog & Feed Management: Processes CRUD operations, enforcing strict separation between B2C permanent catalogs and P2P temporary feeds.
- 3.0 Transaction & Order Processing: Validates shop inventory, calculates cart totals, and manages the lifecycle of tickets in the orders collection.
- 4.0 Real-Time Negotiation Engine: Bypasses standard REST flows, utilizing direct WebSockets/Firestore reads/writes for zero-latency chat and bidding sync.



Data_flow_diagram
(level 0 context diagram)



Data_flow_diagram
(level 1 process breakdown diagram)

Object-Oriented Class Mapping (Pydantic & Controllers)

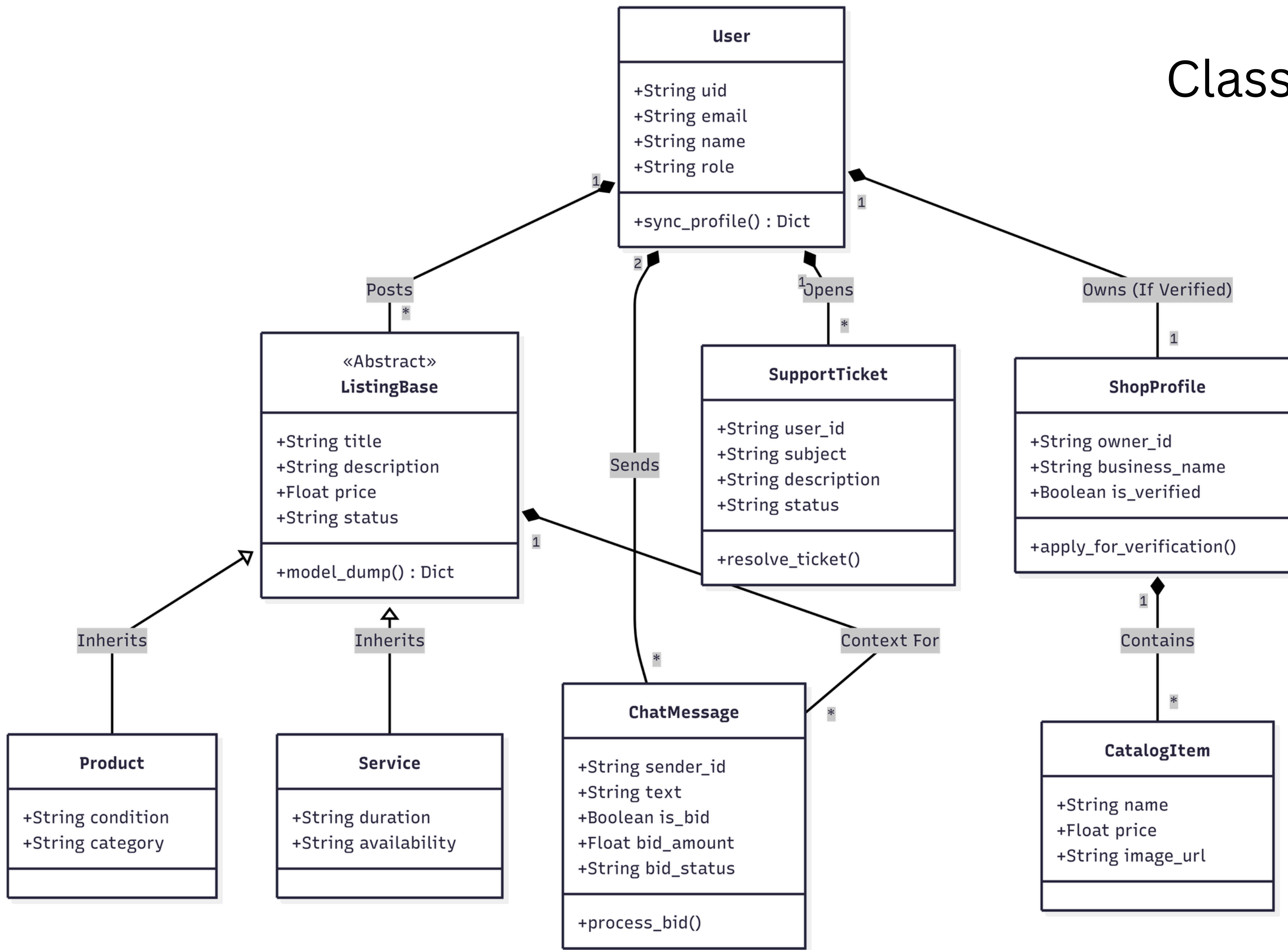
User Hierarchy (Inheritance):

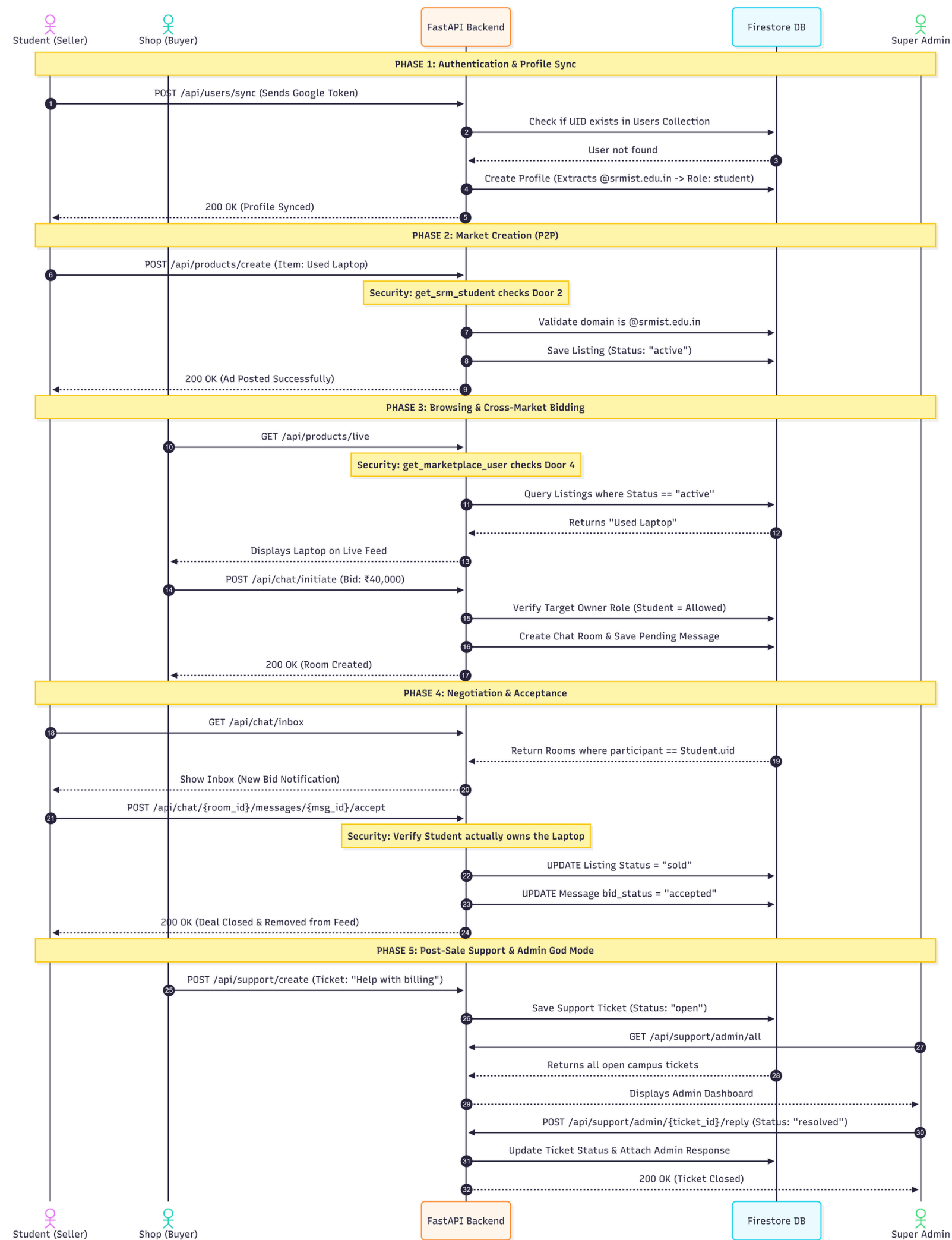
- Account (Base): Holds core attributes (uid, email, role) and authentication methods.
- Student (Child): Adds P2P specific methods (createP2PListing(), checkoutCart(), initiateBid()).
- Shop (Child): Adds B2C vendor attributes (businessName, isVerified) and catalog/order management methods.
- Admin (Child): Adds root-level moderation methods (approveShopRole(), suspendAccount()).

Economy Classes:

- Listing: Manages universal item data (title, price, type, status) and state changes (markAsSold()).
- Order: Manages B2C transaction lifecycles (totalAmount, orderStatus) and inventory validation logic.
- NegotiationRoom: Drives the real-time chat/bidding state (latestBid, isActive) and messaging broadcasts.

Class_diagram(oops)





Communication Flow (B2C Order Execution)

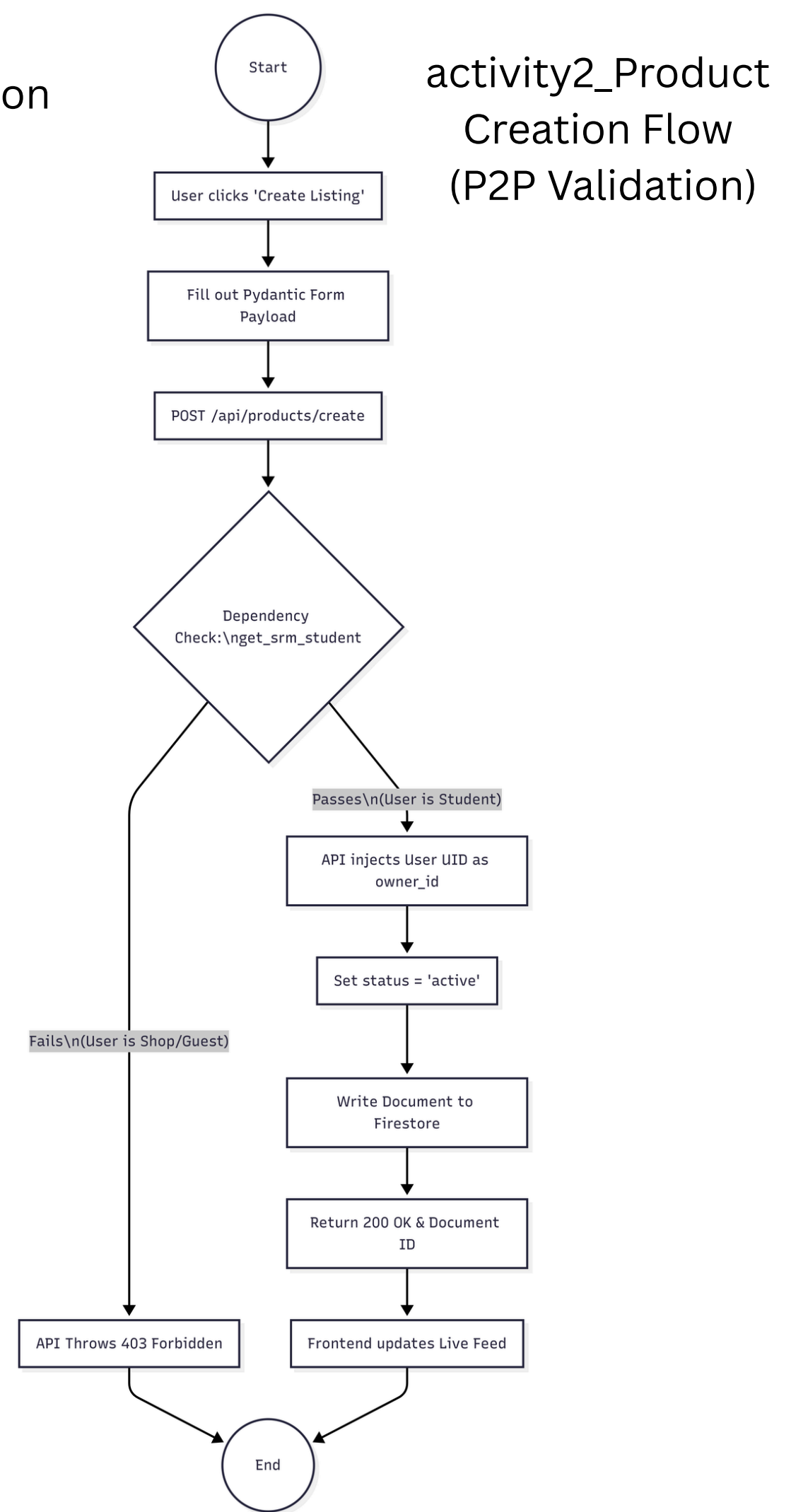
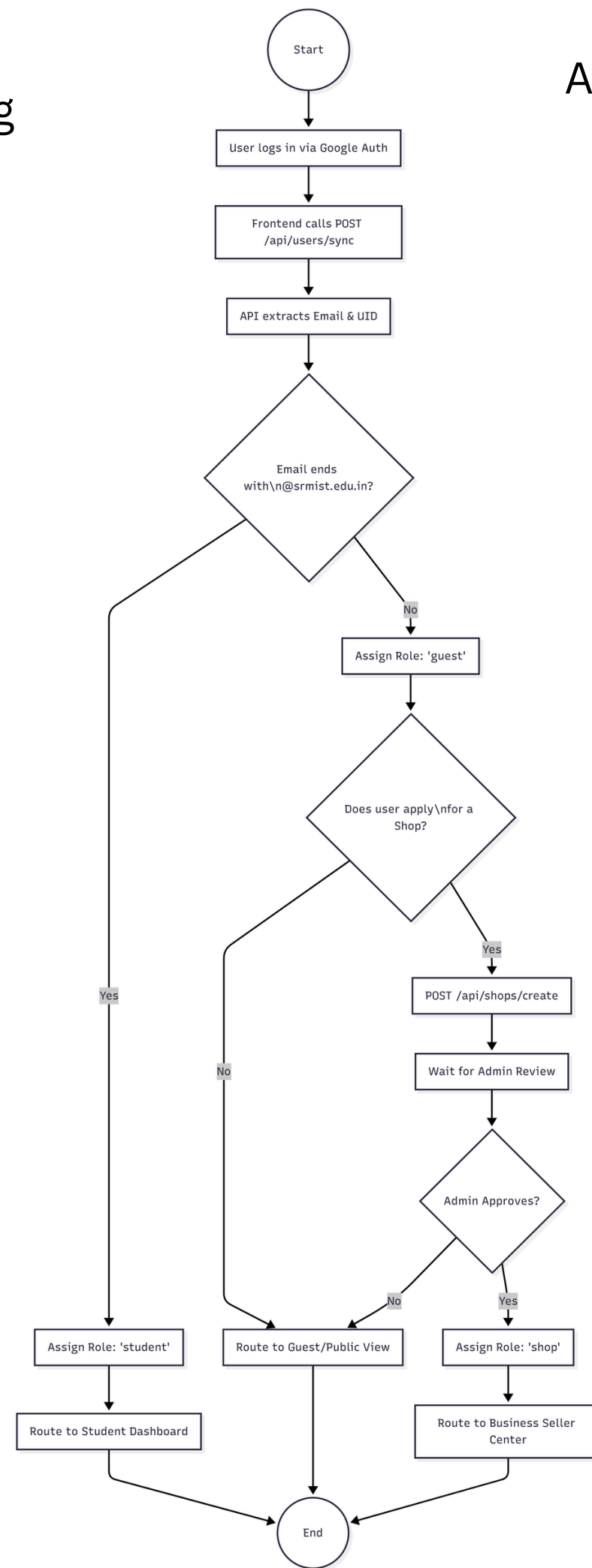
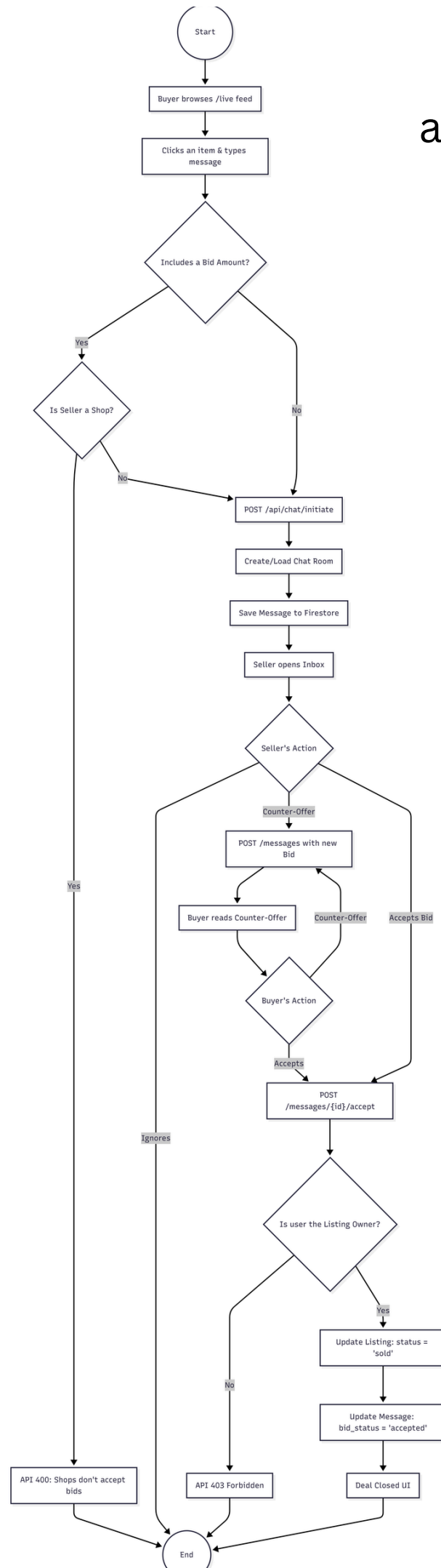
Scenario: A Student placing an order with a Campus Shop

- 1. Student Client: Initiates the transaction by sending an HTTP POST checkout payload to the API Gateway.
- 2. FastAPI & Security ("Student Door"): Intercepts the request and explicitly validates the @srmist.edu.in authentication token before proceeding.
- 3. Order Controller: Receives the secure payload and queries the database to verify shop inventory.
- 4. Firebase Database: Confirms stock availability and executes the createDocument command to generate the order ticket.
- 5. Shop Dashboard (Real-Time): Instantly detects the new database document via Firestore WebSockets, triggering a live "New Order" UI notification that completely bypasses the FastAPI server.

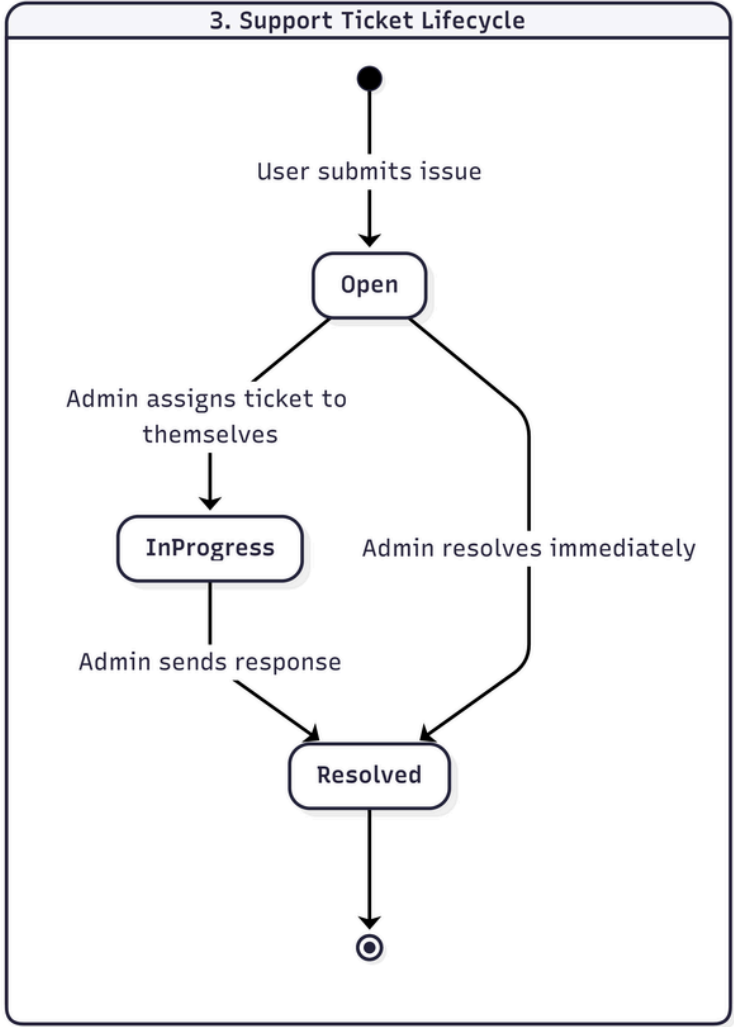
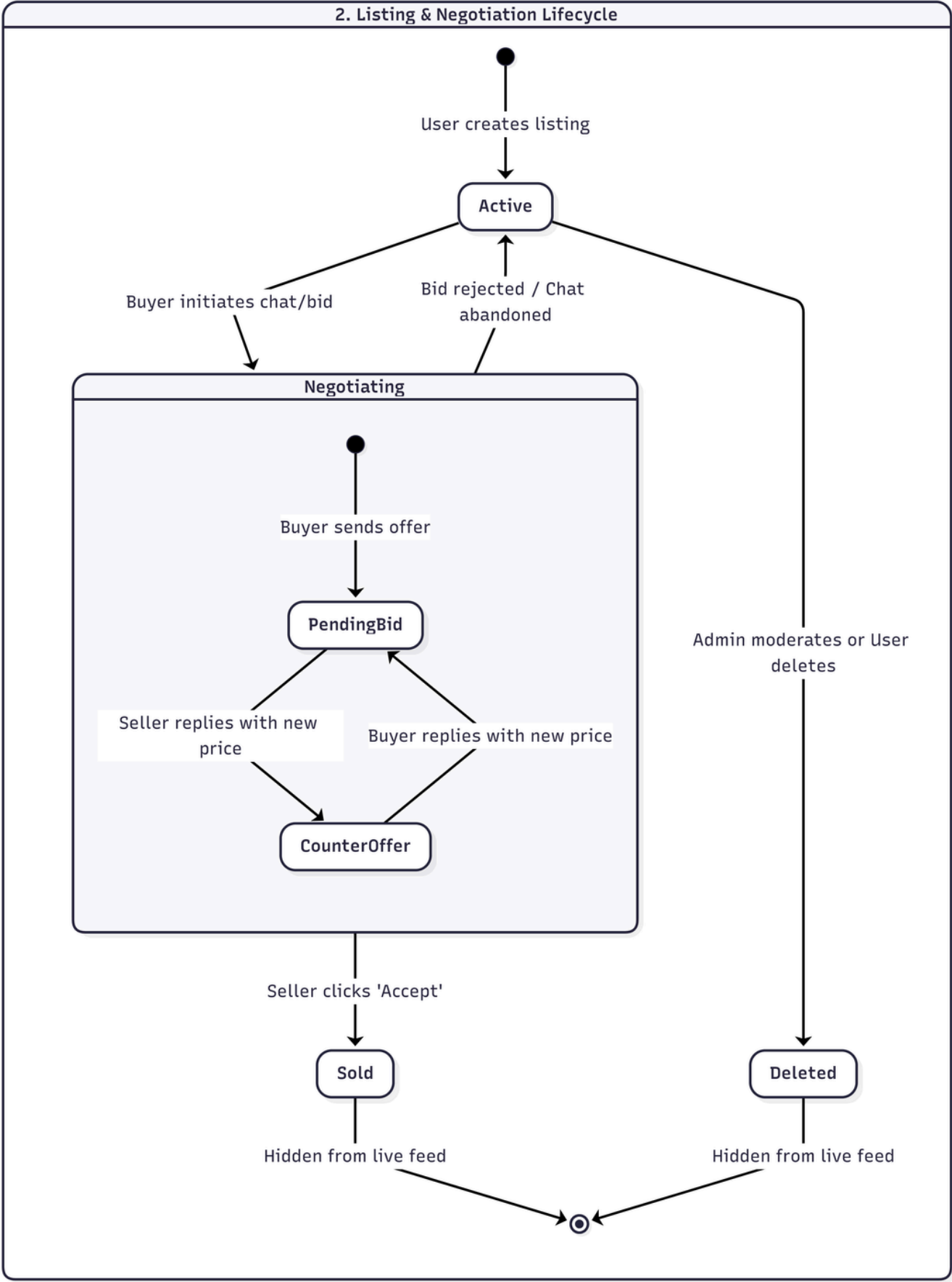
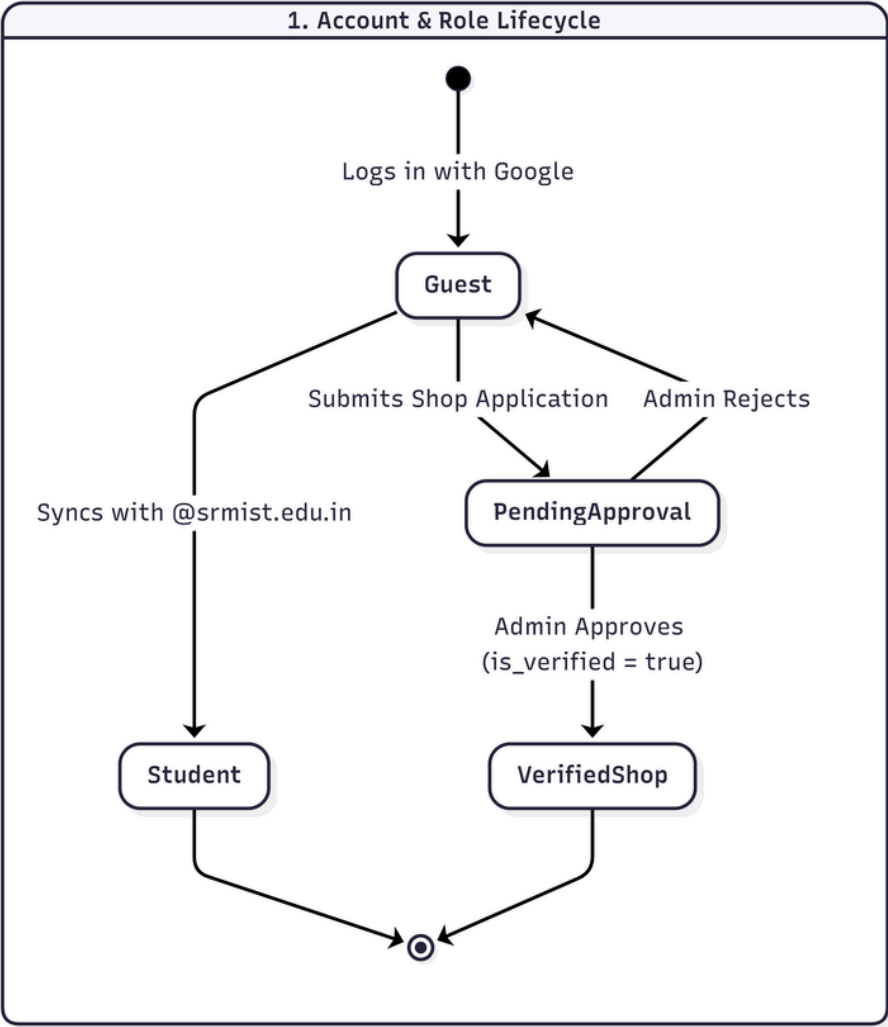
Activity Flow (Shop Verification Process)

Swimlane Breakdown: Role Elevation & Catalog Access

1. Vendor Initiation (Client UI): Vendor logs in as a standard user, is denied dashboard access, and is redirected to submit a Business Verification Form.
2. System Logging (FastAPI / Firebase): The backend registers a pending_verification ticket and places the vendor in a temporary UI "Waiting" state.
3. Administrator Review (God Mode): Admin accesses the pending queue and makes a decision based on campus guidelines (Approve or Reject).
4. Role Elevation (System Execution): Upon approval, the system executes a root-level update, officially changing the vendor's database role from "guest" to "shop".
5. Dashboard Unlock (Final Resolution): The vendor's client detects the new role tag, safely passes the "Shop Door" security dependency, and unlocks access to upload permanent catalog items.



master_state_diagram



Deployment Architecture

Node 1: User Device (Browser / Mobile)

Executes: React/Next.js or Flutter client applications.

Protocols: HTTPS (REST API) and WSS (Secure WebSockets).

Node 2: Cloud Application Server (Docker / VM)

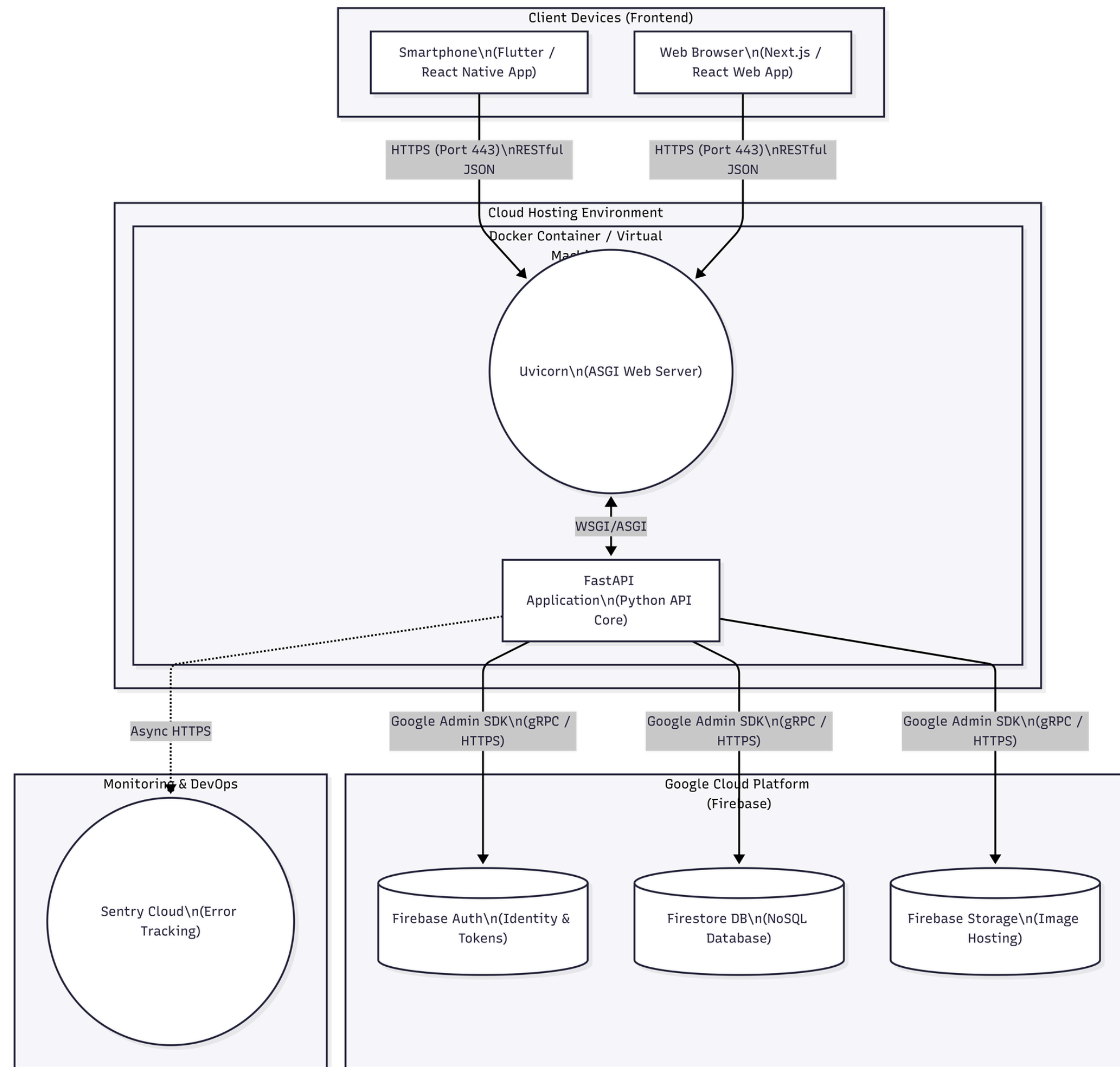
Executes: FastAPI backend via Uvicorn.

Integrations: Sentry for crash reporting; Firebase Admin SDK for secure database communication.

Node 3: Database & Auth (Google Cloud / Firebase)

Executes: Firestore NoSQL Database and Google Identity Platform.

Protocols: Pushes real-time data directly to User Devices via WebSockets (WSS).



Deployment_diagram

Campus Shops

Verified businesses and services around SRM.

CA

VERIFIED

campus cafe
selling the fresh

H block

TE

VERIFIED

TechFix SRM
people

h block

Admin Control Center

Manage shops, users, and support tickets.

- Pending Shops
- Users
- Support

- 📬 Active Inbox
- 🔍 Audit Log (Resolved)

- 📄 All Active
- 🔴 Needs Reply
- ⚠️ Moderation & Appeals
- 💬 General Support

🕒 NEEDS REPLY

🔓 OPEN

📖 CATALOG RULES

Reply Now →

🚩 **REPORT: used books**

Hi! I saw your listing for "used books". Is this still available?

ID Tracker: NNJOYLWATPBMIQZAhODAN

🕒 NEEDS REPLY

🔓 OPEN

👤 USER RULES

🔄 RESTORE

Reply Now →

⚠️ **Official Warning: Policy Violation**

ID Tracker: tFCmXhdxvETLxVYr1JKXeuvebupa2

Campus Live Feed

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All Feed

Products

Services

Requests

₹50000



laptop

ELECTRONICS

good quality and appealing

Campus Member

Message

₹0



used books

BOOKS

good condition , anyone can take for free

Campus Member

REPORT

Message

₹300



used books

BOOKS

good condition

Campus Member

Message

₹1500

₹300

₹650

S

Welcome Back

Log in to manage your campus deals.

Continue with Google

Continue with GitHub

OR

EMAIL ADDRESS

you@example.com

PASSWORD

.....

Log In

Don't have an account yet? [Sign up](#)

S

SRM Market

Support

9+Shops

+ Post

H

Admin Panel

Inbox

Manage your negotiations and campus deals.

Buying (5)

Selling (2)

Support (59)

REPORT: used books

OPEN

Hi! I saw your listing for "used books". Is this still available?

06/04/2026

REPORT: Erodiv Physics Book

RESOLVED

Hi! I saw your listing for "Erodiv Physics Book". Is this still available?

06/04/2026

LISTING RESTORED: 'Crompton Iron'

RESOLVED

Message: I am restoring it

04/04/2026

Official Warning: Policy Violation

OPEN

Bid: ₹

04/04/2026

SUPPORT: please unban me

OPEN

Post a Listing

Cancel

Sell an item, offer a service, or request something on campus.

What are you selling?

e.g., Used Engineering Textbook

Listing Type

Physical Product

Category

Books & Notes

Price (₹)

₹ 0

Description

Describe the details, condition, and any other relevant info...

Post to Live Feed